

# SAME ALGEBRA, DIFFERENT OSCILLATOR: QUANTIZING THE CONE'S POSITION SPINOR

JEREMY EBERT

ABSTRACT. [5] shows the planar Kepler position spinor  $u$  satisfies  $u''(s) + \omega^2 u(s) = 0$  in fictitious time,  $\omega = \frac{1}{2}\sqrt{\mu/T}$ , and notes this is the classical shadow of the established Bohlin/Levi-Civita correspondence between the planar Coulomb problem and a 2D isotropic harmonic oscillator. It leaves open whether quantizing  $u$  reproduces the two-oscillator  $(a_1, a_2)$  system already used for the cone's *shape* coordinates in [4], where  $T = (N + 1)/2$  is *linear* in the total occupation number  $N = n_1 + n_2$ . We resolve this by working the quantum correspondence through explicitly: substituting  $r = \rho^2$  in the planar Coulomb radial Schrödinger equation converts it exactly into a 2D radial oscillator equation, with the oscillator's own energy eigenvalue fixed to (a multiple of) the Coulomb coupling constant and its frequency built from the sought Coulomb energy  $E$  — the roles of energy and coupling constant swap between the two problems. Solving through gives  $E_n = -\alpha^2/(2n - 1)^2$ ,  $n = n_r + |m_\phi| + 1$ , verified by direct numerical solution of the original radial equation (matching to four significant figures, with the degenerate pair  $(n_r, m_\phi) = (1, 0), (0, 1)$  agreeing to  $3 \times 10^{-5}$ ). Since the oscillator quantum number enters through  $\omega \propto \sqrt{-E}$  rather than  $E$  itself,  $T \propto 1/|E|$  is *quadratic*, not linear, in the position-oscillator's quantum number  $N_u := 2n_r + 2|m_\phi|$ . This is a definite, derived answer to the open question:  $N_u \neq N$ , and quantizing the position spinor does not reproduce the shape oscillator of [4] — they are two distinct oscillators built from the same abstract 2D-isotropic-oscillator/ $SU(2)$  mathematics, describing different physical data.

## 1. THE OPEN QUESTION

[5] identified the planar Coulomb problem's classical Levi-Civita regularization with a 2D harmonic oscillator in fictitious time, and flagged — without resolving — the tempting possibility that quantizing this oscillator would simply hand back the same two-mode Schwinger-boson system  $(a_1, a_2)$  already used in [4] for the cone's shape coordinates  $(X, Y, T)$  [1], since a quantized 2D isotropic oscillator generically carries an  $SU(2)$  degeneracy structure built the same way. A quick scaling check argued against this: [4]'s  $T = (N + 1)/2$  is linear in  $N = n_1 + n_2$ , while the Bohr-type correspondence  $T \propto n^2$  expected for a Coulomb problem suggested the position-spinor's own oscillator quantum number would need to scale quadratically with the physical principal quantum number — a mismatch worth resolving properly rather than leaving as a suspicion.

## 2. THE CORRESPONDENCE, WORKED THROUGH

Write the planar (2D) Coulomb radial Schrödinger equation, angular momentum quantum number  $m_\phi \in \mathbb{Z}$ , coupling  $\alpha > 0$ , in the convention  $\hbar = 1$ , mass  $= \frac{1}{2}$ :

$$(1) \quad -R''(r) - \frac{1}{r}R'(r) + \frac{m_\phi^2}{r^2}R(r) - \frac{\alpha}{r}R(r) = ER(r).$$

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**Theorem 2.1** (The Bohlin substitution converts (1) into a 2D oscillator equation). *Setting  $r = \rho^2$  and  $R(r) =: \mathcal{R}(\rho)$ ,*

$$(2) \quad -\mathcal{R}''(\rho) - \frac{1}{\rho}\mathcal{R}'(\rho) + \frac{4m_\phi^2}{\rho^2}\mathcal{R}(\rho) - 4E\rho^2\mathcal{R}(\rho) = 4\alpha\mathcal{R}(\rho).$$

*This is exactly the radial equation of a 2D isotropic harmonic oscillator with angular momentum  $L = 2m_\phi$  and frequency  $\omega = 2\sqrt{-E}$ , whose own energy eigenvalue is fixed at  $\mathcal{E} = 4\alpha$ .*

*Proof.* Direct chain-rule substitution ( $d/dr = (1/2\rho)d/d\rho$ ), carried out symbolically, converting (1) into (2) with zero residual after clearing denominators by  $4\rho^2$ . Equation (2) is then read off by comparison with the standard 2D radial oscillator equation  $-\chi'' - \frac{1}{\rho}\chi' + \frac{L^2}{\rho^2}\chi + \omega^2\rho^2\chi = \mathcal{E}\chi$ .  $\square$

*Remark.* The mechanism is the same one used for the 3D Coulomb–oscillator correspondence via the Kustaanheimo–Stiefel transformation [6]: the physical Coulomb energy  $E$ , the quantity being sought, becomes the oscillator’s *frequency*, while the Coulomb coupling constant  $\alpha$ , a fixed input, becomes the oscillator’s *energy eigenvalue*. This swap of roles is exactly why the resulting spectrum is quadratically, not linearly, suppressed in the oscillator’s own quantum number: the standard 2D oscillator spectrum is  $\mathcal{E} = \omega(2n_r + |L| + 1)$ , *linear* in  $(2n_r + |L|)$ , but here  $\mathcal{E} = 4\alpha$  is fixed and  $\omega = 2\sqrt{-E}$  is being solved for, so it is  $\sqrt{-E}$ , not  $E$ , that comes out linear in the oscillator quantum number.

**Corollary 2.2** (The 2D hydrogen spectrum).

$$(3) \quad E_n = -\frac{\alpha^2}{(2n-1)^2}, \quad n := n_r + |m_\phi| + 1 = 1, 2, 3, \dots$$

*Proof.* Substitute  $L = 2m_\phi$ ,  $\omega = 2\sqrt{-E}$  into  $\mathcal{E} = \omega(2n_r + |L| + 1) = 4\alpha$ :  $2\sqrt{-E}(2n_r + 2|m_\phi| + 1) = 4\alpha$ , so  $\sqrt{-E} = 2\alpha/(2n_r + 2|m_\phi| + 1) = 2\alpha/(2n-1)$ , giving (3).  $\square$

**Numerical Verification.** Solving (1) directly by a shooting method ( $\alpha = 1.3$ ,  $\alpha^2 = 1.69$ ), independent of the oscillator substitution above:  $(n_r, m_\phi) = (0, 0)$  [ $n = 1$ ] gives  $E = -1.68912$  against  $-\alpha^2/(2 \cdot 1 - 1)^2 = -1.69$ ;  $(n_r, m_\phi) = (1, 0)$  and  $(0, 1)$  [both  $n = 2$ ] give  $E = -0.18775$  and  $-0.18778$  respectively, against  $-\alpha^2/(2 \cdot 2 - 1)^2 = -0.18778$ , and agree with each other to  $3 \times 10^{-5}$ , confirming the expected degeneracy (the hidden  $SO(3)$  symmetry of the 2D hydrogen atom, distinct from the full  $SO(4)$  of the 3D case). All three numbers match (3) to four significant figures.

### 3. RESOLVING THE OPEN QUESTION

Write  $N_u := 2n_r + 2|m_\phi|$  for the position-spinor oscillator’s own quantum number, so (3) reads  $\sqrt{-E} = 2\alpha/(N_u + 1)$ , i.e.

$$(4) \quad E \propto -\frac{1}{(N_u + 1)^2}.$$

Since the classical cone coordinate is  $T \propto 1/|E|$  (Section 3 of [5]:  $\omega = \frac{1}{2}\sqrt{\mu/T}$  gives  $T \propto 1/\omega^2 \propto 1/|E|$  under the identification  $\omega \leftrightarrow \sqrt{-E}$ ), (4) gives

$$(5) \quad T \propto (N_u + 1)^2 \quad (\text{quadratic in } N_u),$$

in direct contrast to [4]’s  $T = (N + 1)/2$  (linear in the shape-oscillator’s  $N = n_1 + n_2$ ).

**Corollary 3.1** (The two oscillators are provably distinct).  $N_u \neq N$ : *there is no consistent identification of the fictitious-time position oscillator’s quantum number with the shape oscillator’s quantum number of [4], since the two obey different (quadratic vs. linear) relations to the same coordinate  $T$ . Quantizing the Levi-Civita position spinor  $u$  does not reproduce the shape oscillator  $(a_1, a_2)$ ; it is a different oscillator, built from the same abstract mathematics (a 2D isotropic oscillator carries an  $SU(2)$  degeneracy structure regardless of what it physically represents), describing different physical data — position-in-fictitious-time versus abstract orbital shape.*

*Remark* (What remains open). This resolves the specific scaling question but not the broader one: whether the two oscillators combine into some larger structure, or relate by some map more subtle than direct identification of quantum numbers. We have not looked for one. Nor have we addressed the genuine technical subtlety flagged in [5] — that quantizing a fictitious-time-reparametrized system correctly (in the sense of the Duru–Kleinert path-integral treatment) requires care beyond naively quantizing the classical  $u'' + \omega^2 u = 0$  equation. The derivation here works directly from the time-independent Schrödinger equation for the physical (real-time) planar Coulomb problem, which sidesteps that subtlety rather than resolving it: we have shown what the correct quantum spectrum is and how it relates to the oscillator language, but we have not derived it from the fictitious-time formalism itself.

#### AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic’s Claude) were used in the derivation, symbolic and numerical cross-checking, and L<sup>A</sup>T<sub>E</sub>X preparation of this note. This includes: symbolic (computer-algebra) verification of the substitution in Theorem 2.1; numerical solution (shooting method) of the original radial Schrödinger equation (1) to verify Corollary 2.2 independently of the oscillator-correspondence derivation, including a degeneracy check between two different  $(n_r, m_\phi)$  pairs; and identification and correction of an initial factor-of-four arithmetic error in simplifying the energy formula to its  $(2n - 1)$  form, caught by comparing against the direct numerical solution. All mathematical claims were independently verified by the author.

#### REFERENCES

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FRANKLIN COUNTY, OHIO, USA